

ANOVA 2

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Chapters for ANOVA/ANCOVA

R book

Chapter 11: Analysis of Variance and

Chapter 12: Analysis of Covariance

ISLR

Not covered per se, but you can check for relevant materials

Chapter 3, Section 3.3.1: Qualitative Predictors

Last time: two parametrizations

```
mod1 <- lm(yield~soil, data=yields, contrasts = "contr.sum")  
round(summary(mod1)$coef,4)
```

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	11.9	0.6241	19.0673	0.0000
##	soil(clay)	-0.4	0.8826	-0.4532	0.6540
##	soil(loam)	2.4	0.8826	2.7192	0.0113

```
mod2 <- lm(yield~soil, data=yields, contrasts = "contr.treatment")  
round(summary(mod2)$coef,4)
```

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	11.5	1.0810	10.6385	0.0000
##	soil(loam)	2.8	1.5287	1.8316	0.0781
##	soil(sand)	-1.6	1.5287	-1.0466	0.3046

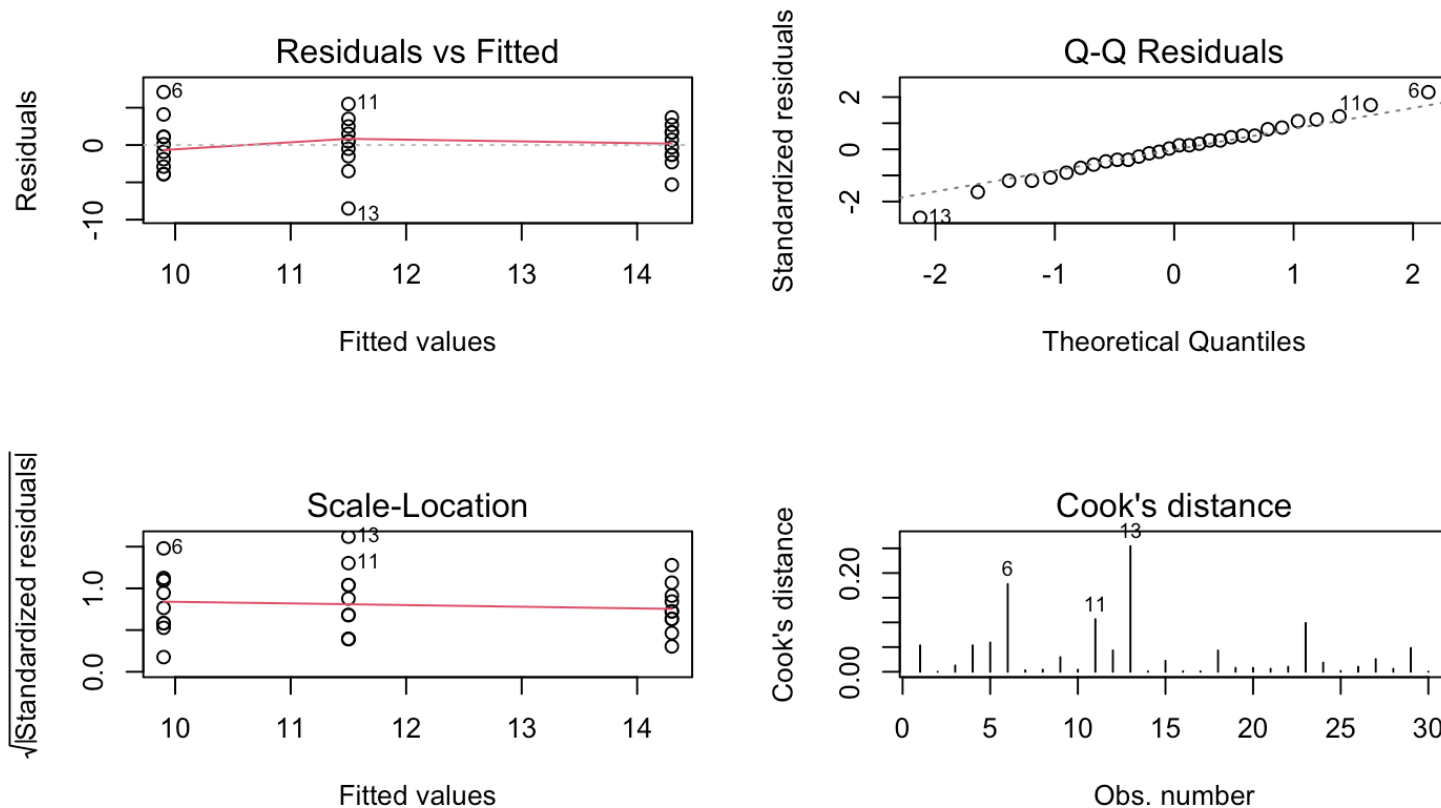
anova(mod1) and anova(mod2)

```
## Analysis of Variance Table
##
## Response: yield
##           Df Sum Sq Mean Sq F value Pr(>F)
## soil        2   99.2   49.600   4.2447 0.02495 *
## Residuals  27  315.5   11.685
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Analysis of Variance Table
##
## Response: yield
##           Df Sum Sq Mean Sq F value Pr(>F)
## soil        2   99.2   49.600   4.2447 0.02495 *
## Residuals  27  315.5   11.685
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual analysis

We should also for ANOVA-models check the assumptions by residual analysis



ANOVA with several factors

- Each with two or more levels
- Assume two factors: A with $i = 1, 2, \dots, a$ levels, and B with $j = 1, 2, \dots, b$ levels
- If all levels of A are tested with all levels of B we have a complete crossed design
- Assume there are $k = 1, 2, \dots, r$ replicates of all ab level combinations so
- So, the total number of observations is $n = a \cdot b \cdot r$

Illustration of two-factor case

<https://solve.shinyapps.io/Factorialdemo/>

The Model

Model:

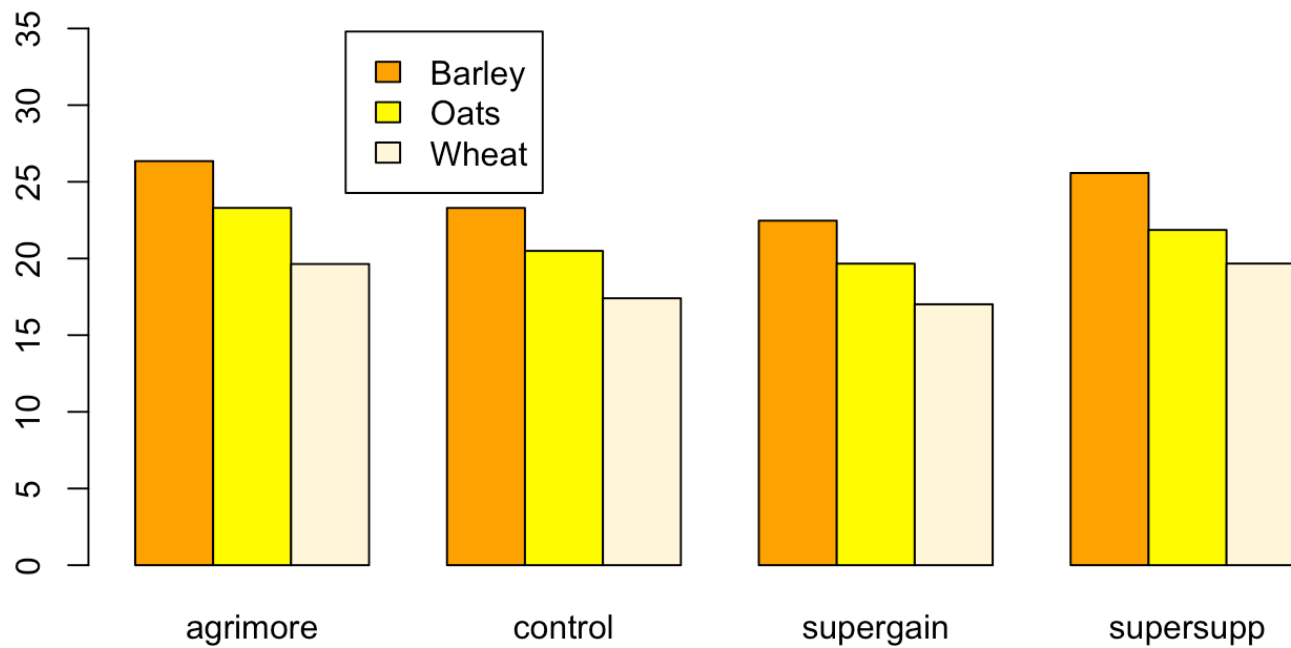
$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

where $\epsilon_{ijk} \sim N(0, \sigma^2)$ and $(\alpha\beta)_{ij}$ is the effect due to any interaction between the i -th level of and the j -th level of.

Parametrization (restrictions)

- Sum-to-zero: $\sum_i \alpha_i = 0$, $\sum_j \beta_j = 0$, $\sum_i (\alpha\beta)_{ij} = 0$ and $\sum_j (\alpha\beta)_{ij} = 0$.
- Reference levels: $\alpha_1 = 0$, $\beta_1 = 0$, $(\alpha\beta)_{1,j} = 0$ for all j and $(\alpha\beta)_{i,1} = 0$ for all i .

Example: Animal diets - “growth data” from R book



R structure of the data

```
##      supplement  diet      gain
## 1    supergain  wheat 17.37125
## 2    supergain  wheat 16.81489
## 3    supergain  wheat 18.08184
## 4    supergain  wheat 15.78175
## 5      control  wheat 17.70656
## 6      control  wheat 18.22717
## 7      control  wheat 16.08650
## 8      control  wheat 17.60184
## 9    supersupp  wheat 20.78861
## 10   supersupp  wheat 20.00858
## 11   supersupp  wheat 19.30287
## 12   supersupp  wheat 18.57330
## 13   agrimore   wheat 19.54688
## 14   agrimore   wheat 17.96440
## 15   agrimore   wheat 21.43738
## 16   agrimore   wheat 19.60763
```

R analysis

For reference level analysis, see “the R-book” on page 517.

Let’s again use the sum-to-zero parametrization here:

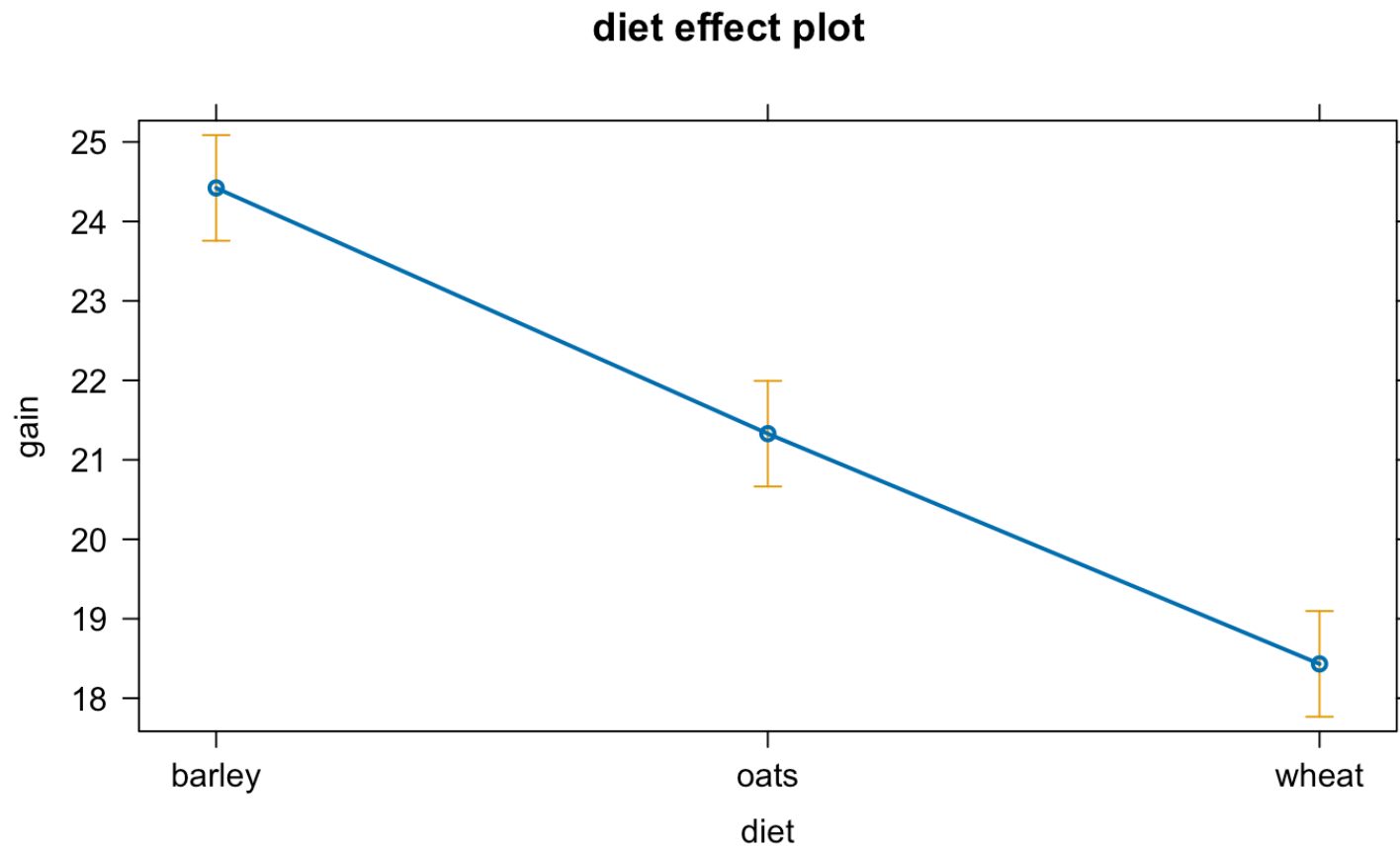
```
mod2 <- mixlm::lm(gain ~ diet + supplement + diet:supplement,  
  data=weights, contrasts = "contr.sum")
```

```
options(contrasts = c("contr.sum", "contr.poly"))  
mod2 <- stats::lm(gain ~ diet + supplement + diet:supplement,  
  data=weights)
```

Main effects plot of **Diet**

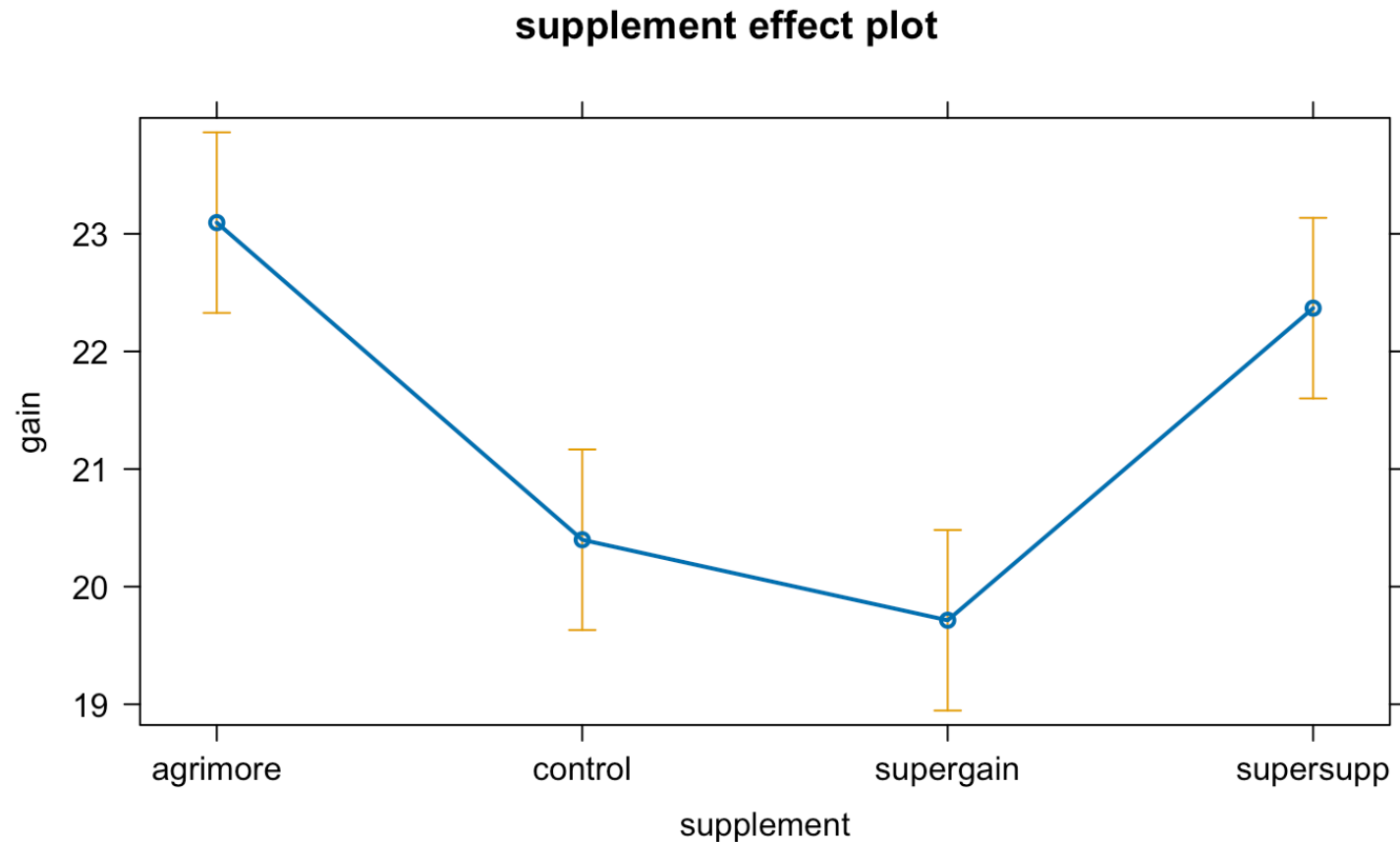
“Main effect plots” of group means with 95% CI's

```
library(effects)  
plot(Effect(focal.predictors = c("diet"), mod = mod2))
```



Main effects plot of Supplement

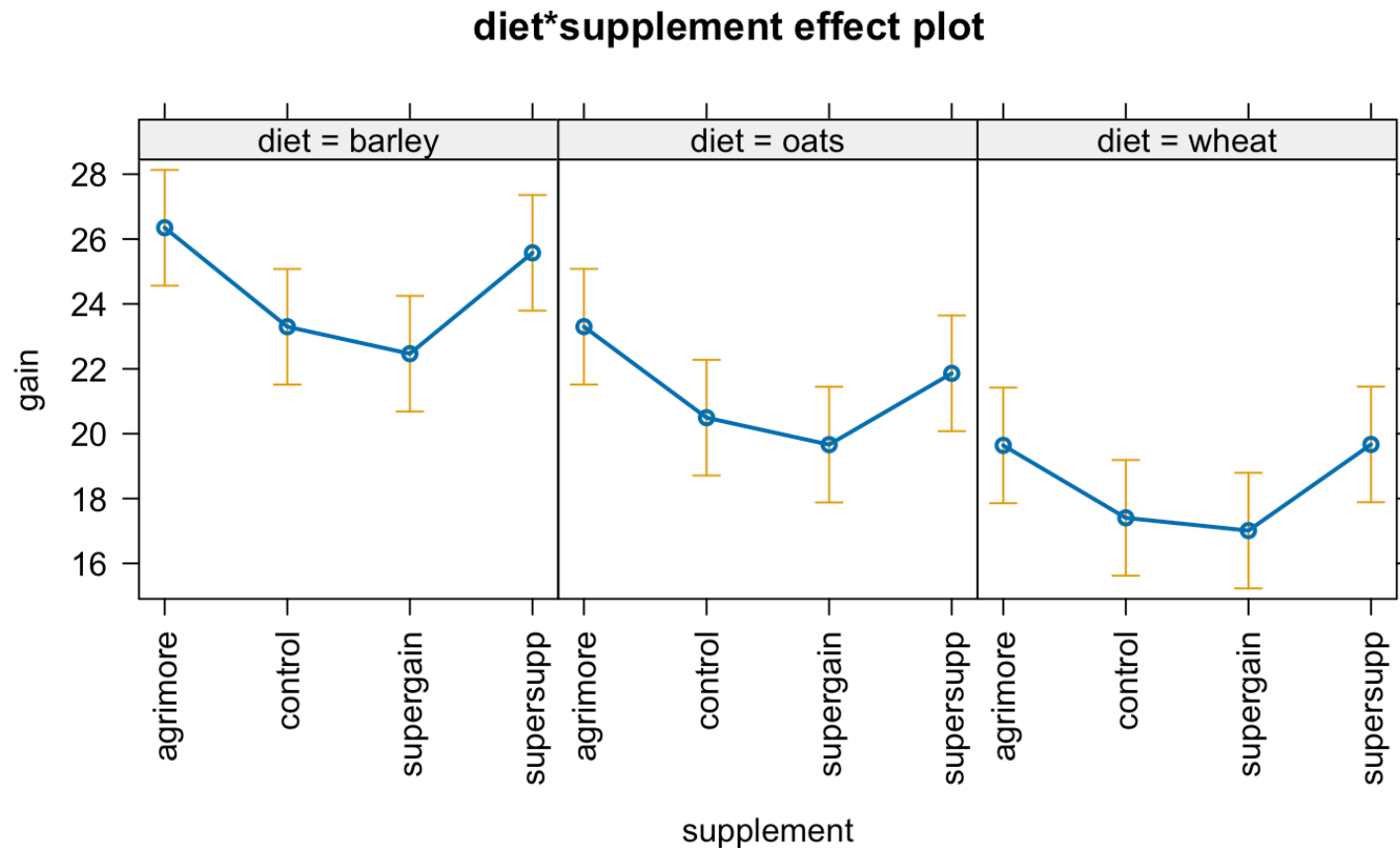
```
plot(Effect(c("supplement"), mod2))
```



Interaction plots

Means of diet:supplement combinations.

```
plot(allEffects(mod2, confidence.level=.99), layout=c(3,1),  
     rotx=90, alternating=FALSE)
```



Model parameter estimation

Still maximum likelihood maximization/ SSE minimization in the regression formulation of 2-way ANOVA with interactions to get $\widehat{\beta}_0, \widehat{\beta}_1, \dots, \widehat{\beta}_p$ which can similarly to how we did it last time translated into $\widehat{\mu}, \widehat{\alpha}_i, \widehat{\beta}_j, \widehat{(\alpha\beta)}_{ij}$. For balanced data the least squares estimators for the model parameters are (sum-to-zero):

$$\widehat{\mu} = \bar{y}_{...}$$

$$\widehat{\alpha}_i = \bar{y}_{i..} - \bar{y}_{...}$$

$$\widehat{\beta}_j = \bar{y}_{.j.} - \bar{y}_{...}$$

$$\widehat{(\alpha\beta)}_{ij} = \bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}$$

Interpretation: effects are **deviations** from the overall mean μ .

Model parameter estimation

Now take level 1 of A and B as reference

$$\hat{\mu}^{(ref)} = \bar{y}_{11}.$$

$$\hat{\alpha}_i^{(ref)} = \bar{y}_{i1.} - \bar{y}_{11.}, \quad i = 2, \dots, a$$

$$\hat{\beta}_j^{(ref)} = \bar{y}_{1j.} - \bar{y}_{11.}, \quad j = 2, \dots, b$$

$$\widehat{(\alpha\beta)}_{ij}^{(ref)} = \bar{y}_{ij.} - \bar{y}_{i1.} - \bar{y}_{1j.} + \bar{y}_{11.}, \quad i = 2, \dots, a, \quad j = 2, \dots, b$$

Interpretation: effects are **differences** from the reference.

As usual, the noise variance estimate is:

$$\hat{\sigma}^2 = MSE$$

Summary with sum to zero:

Sum to zero parametrization - deviations from the grand mean:

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	21.3939	0.1893	113.0450	0.0000
## diet(barley)	3.0277	0.2676	11.3125	0.0000
## diet(oats)	-0.0651	0.2676	-0.2433	0.8092
## supplement(agrmore)	1.7014	0.3278	5.1904	0.0000
## supplement(control)	-0.9953	0.3278	-3.0364	0.0044
## supplement(supergain)	-1.6801	0.3278	-5.1254	0.0000
## diet(barley):supplement(agrmore)	0.2255	0.4636	0.4864	0.6297
## diet(oats):supplement(agrmore)	0.2682	0.4636	0.5785	0.5665
## diet(barley):supplement(control)	-0.1297	0.4636	-0.2797	0.7813
## diet(oats):supplement(control)	0.1602	0.4636	0.3455	0.7317
## diet(barley):supplement(supergain)	-0.2754	0.4636	-0.5942	0.5561
## diet(oats):supplement(supergain)	0.0143	0.4636	0.0308	0.9756

Summary with reference:

Reference diet: **barley**, supplement: **agrimore**

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	26.3485	0.6556	40.1907	0.0000
## diet(oats)	-3.0501	0.9271	-3.2898	0.0022
## diet(wheat)	-6.7094	0.9271	-7.2367	0.0000
## supplement(control)	-3.0518	0.9271	-3.2917	0.0022
## supplement(supergain)	-3.8824	0.9271	-4.1875	0.0002
## supplement(supersupp)	-0.7732	0.9271	-0.8339	0.4098
## diet(oats):supplement(control)	0.2471	1.3112	0.1885	0.8516
## diet(wheat):supplement(control)	0.8183	1.3112	0.6241	0.5365
## diet(oats):supplement(supergain)	0.2470	1.3112	0.1884	0.8517
## diet(wheat):supplement(supergain)	1.2557	1.3112	0.9577	0.3446
## diet(oats):supplement(supersupp)	-0.6650	1.3112	-0.5072	0.6151
## diet(wheat):supplement(supersupp)	0.8024	1.3112	0.6120	0.5444

We want to test hypotheses

Interaction effects?

$$H_0 : (\alpha\beta)_{11} = (\alpha\beta)_{12} = \cdots = (\alpha\beta)_{ab} = 0$$

Main effect of A?

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_a = 0$$

Main effect of B?

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_b = 0$$

Splitting of sums of squares

The Total Sums of Squares measures total variability:

$$SST = \sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{...})^2$$

Can be split into:

$$SST = SSA + SSB + SSAB + SSE$$

Degrees of freedom follow correspondingly as

$$(n - 1) = [(a - 1)] + [(b - 1)] + [(a - 1)(b - 1)] + [n - ab].$$

This corresponds to a linear regression formulation with

$$SSR = SSA + SSB + SSAB \text{ and } p = ab - 1$$

F-tests

Interaction: Reject H_0 if

$$F = \frac{MSAB}{MSE} = \frac{SSAB/[(a-1)(b-1)]}{SSE/[n-ab]}$$

is larger than $F_{\alpha, (a-1)(b-1), n-ab}$ or p-value $< \alpha$.

Main effect A: Reject H_0 if

$$F = \frac{MSA}{MSE} > F_{\alpha, (a-1), n-ab}$$

Main effect B: Reject H_0 if

$$F = \frac{MSB}{MSE} > F_{\alpha, (b-1), n-ab}$$

The ANOVA-table for the growth data

```
anova(mod2)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: gain
```

##		Df	Sum Sq	Mean Sq	F value	Pr(>F)	
##	diet	2	287.171	143.586	83.5201	2.999e-14	***
##	supplement	3	91.881	30.627	17.8150	2.952e-07	***
##	diet:supplement	6	3.406	0.568	0.3302	0.9166	
##	Residuals	36	61.890	1.719			

```
## ----
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusions?

A reduced model

```
mod3 <- lm(gain ~ diet + supplement, data=weights, contrasts = "contr.sum")
anova(mod3)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: gain
```

```
##           Df  Sum Sq Mean Sq F value    Pr(>F)
## diet         2 287.171 143.586   92.358 4.199e-16 ***
## supplement   3  91.881  30.627   19.700 3.980e-08 ***
## Residuals   42  65.296   1.555
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note: The complete crossed design gives “orthogonal effects”. The SS’s are independent, and, apart from SSE, they do not change when effects are removed/added to the model.

Tukey tests supplement

```
p-tests <- mixlm::simple.glht(mod3, "supplement", corr = c("Tukey"),  
                             level = 0.95)  
print(round(p-tests$res, 4))
```

##	Lower	Center	Upper	Std.Err	t value	P(>t)
## agrimore-control	1.3351	2.6967	4.0583	0.509	5.2977	0.0000
## agrimore-supergain	2.0198	3.3815	4.7431	0.509	6.6429	0.0000
## agrimore-supersupp	-0.6343	0.7274	2.0890	0.509	1.4289	0.4889
## control-supergain	-0.6769	0.6848	2.0464	0.509	1.3452	0.5400
## control-supersupp	-3.3310	-1.9693	-0.6077	0.509	-3.8688	0.0020
## supergain-supersupp	-4.0157	-2.6541	-1.2925	0.509	-5.2141	0.0000

Tukey tests diet

```
p-tests2 <- mixlm::simple.glht(mod3, "diet", corr = c("Tukey"),  
                             level = 0.95)  
print(round(p-tests2$res,4))
```

##		Lower	Center	Upper	Std.Err	t value	P(>t)
##	barley-oats	2.0218	3.0928	4.1638	0.4408	7.0159	0
##	barley-wheat	4.9193	5.9903	7.0613	0.4408	13.5886	0
##	oats-wheat	1.8265	2.8975	3.9685	0.4408	6.5727	0

Compact letter display

A compact letter display (cld) of groups of effects

```
cld(ptests)
```

```
## Tukey's HSD
```

```
## Alpha: 0.05
```

```
##
```

```
##           Mean G1 G2
```

```
## agrimore  23.09531    B
```

```
## supersupp 22.36796    B
```

```
## control   20.39861    A
```

```
## supergain 19.71385    A
```

Analysis of covariance - ANCOVA

The dataset `birth` in the `BIAS.data` package holds records of 189 birth weights from Massachusetts, USA, and some additional variables. Three of the variables are,

Var	Description
LWT	Weight of mother.
SMK	YES if mother smokes and NO otherwise.
BWT	Birth weight (RESPONSE).

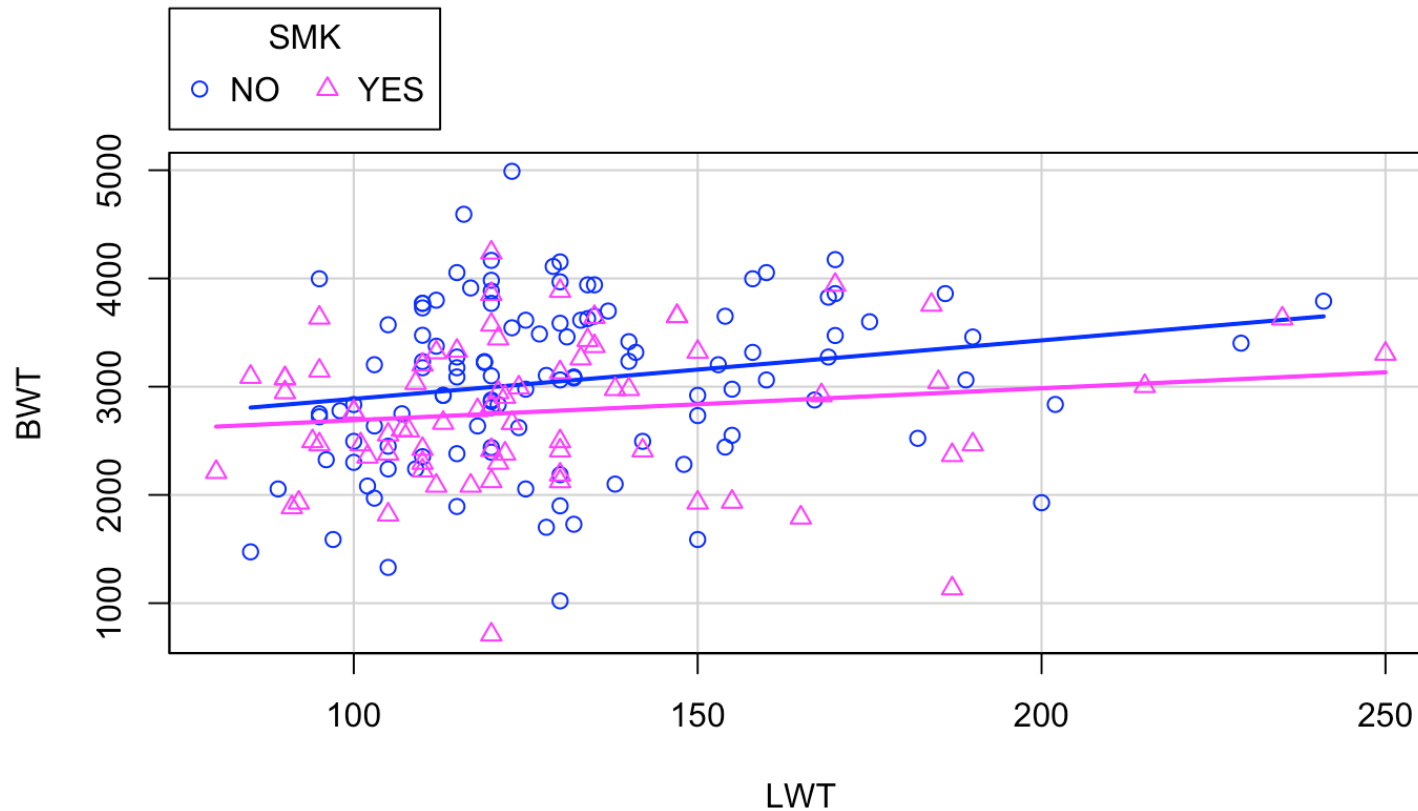
The main question is: Does mother's smoking affect birth weight and does it depend on the mother weight?

We have both a categorical and a continuous predictor for the response variable.

The data

##		LWT	SMK	BWT
##	1	120	YES	709
##	2	130	NO	1021
##	3	187	YES	1135
##	4	105	NO	1330
##	5	85	NO	1474
##	6	150	NO	1588
##	7	97	NO	1588

Scatterplot with separate linear models for smokers and non-smokers



ANCOVA as regression with indicator variable

Let $y = BW T$ and $x_1 = LWT$ and define

$$x_{2i} = \begin{cases} 0 & \text{if smoker} \\ 1 & \text{if non-smoker} \end{cases}$$

Assume the model

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \epsilon_i$$

where $\epsilon_i \sim N(0, \sigma^2)$.

Fitting the model in R

```
birth4 <- lm(BWT ~ LWT + SMK, data=birth, contrasts = "contr.treatment")
summary(birth4)
```

```
##
## Call:
## lm(formula = BWT ~ LWT + SMK, data = birth, contrasts = "contr.treatment")
##
## Residuals:
```

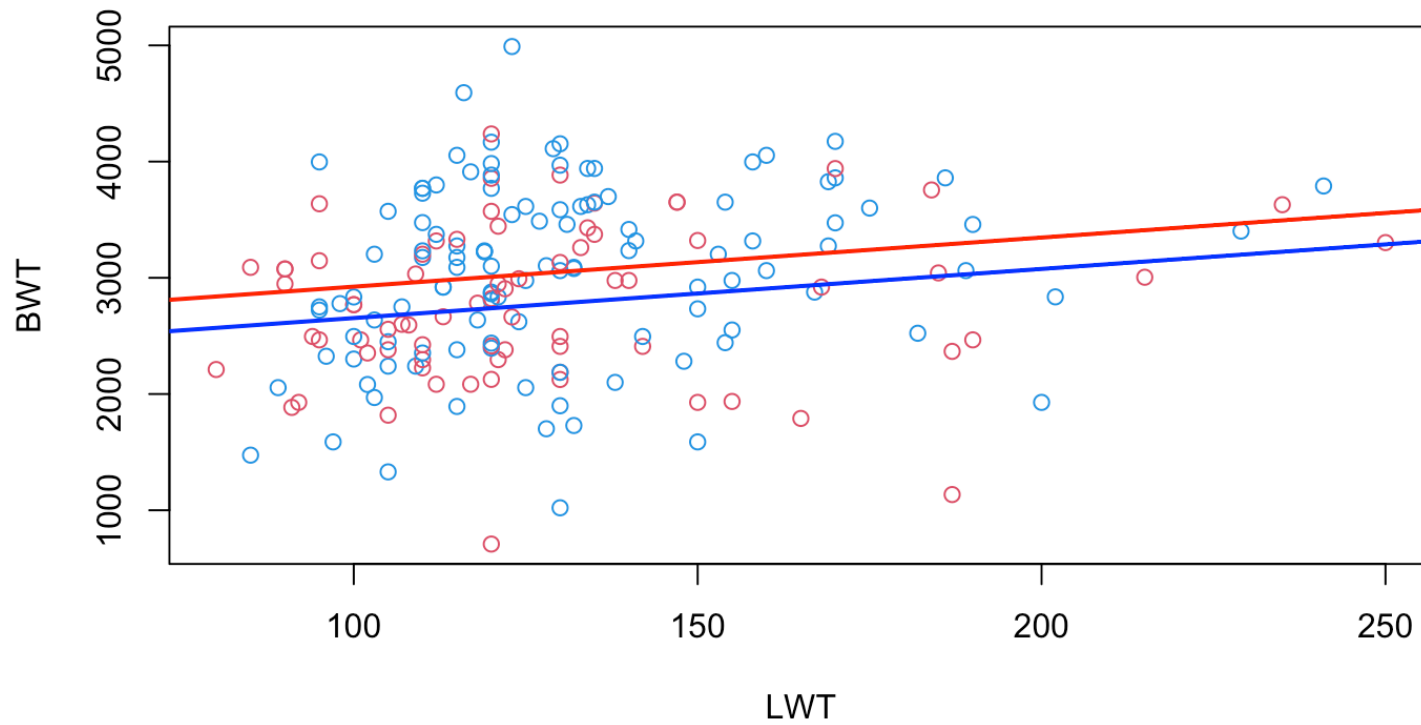
	Min	1Q	Median	3Q	Max
	-2030.16	-447.00	27.74	514.20	1968.51

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2500.174	230.833	10.831	<2e-16 ***
LWT	4.238	1.690	2.508	0.0130 *
SMK(YES)	-270.013	105.590	-2.557	0.0114 *

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## s: 707.8 on 186 degrees of freedom
## Multiple R-squared:  0.06731,
## Adjusted R-squared:  0.05728
```

The estimated model(s)



The model with interaction term

Assume that we expand the model with an interaction term between LWT and SMK

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} \cdot x_{2i} + \epsilon_i$$

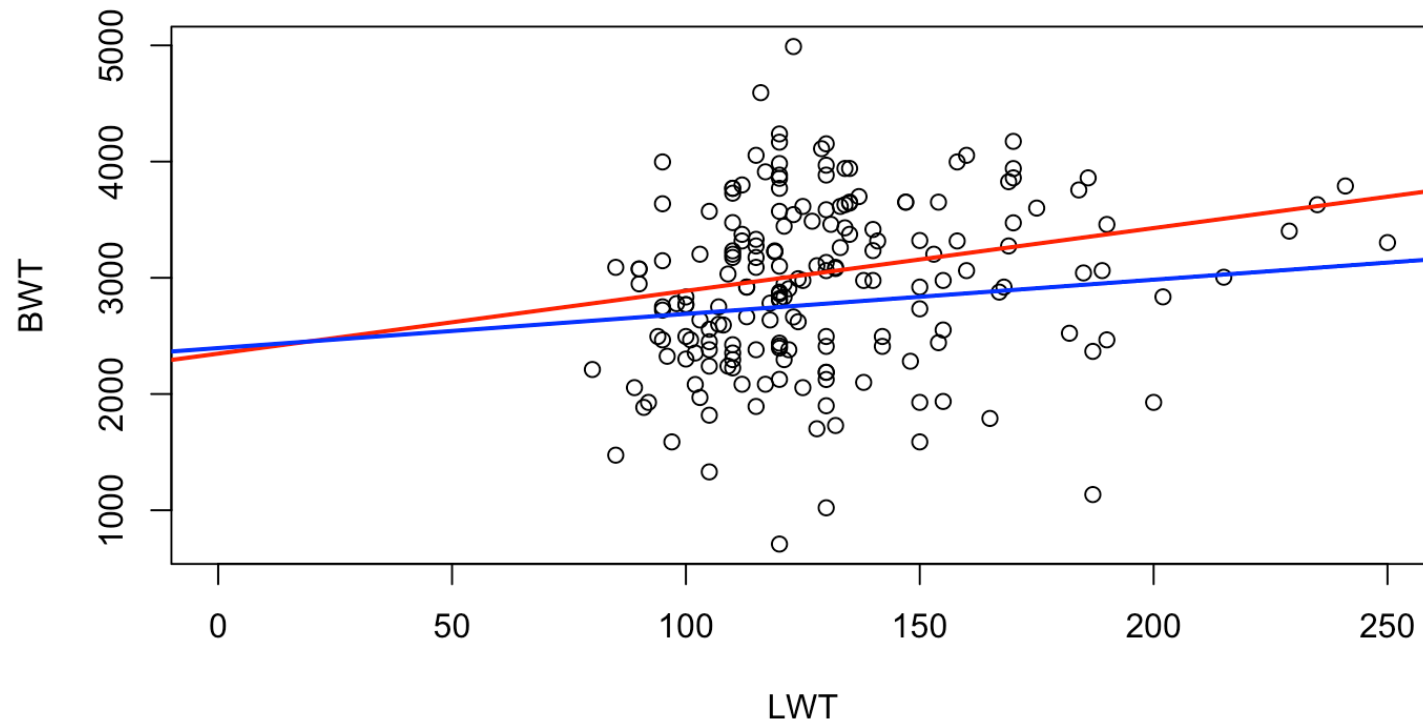
with $\epsilon_i \sim N(0, \sigma^2)$.

Lets see how the model is for different values of x_{2i}

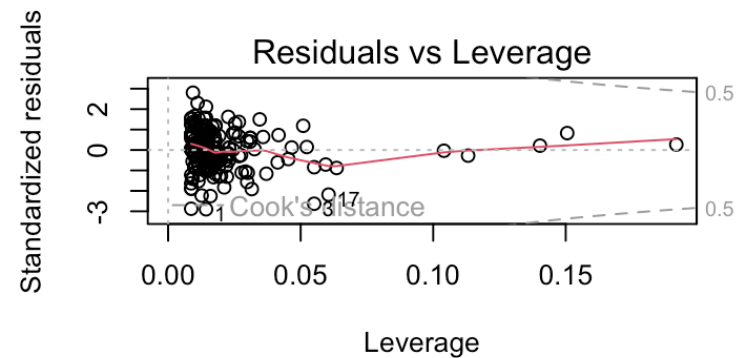
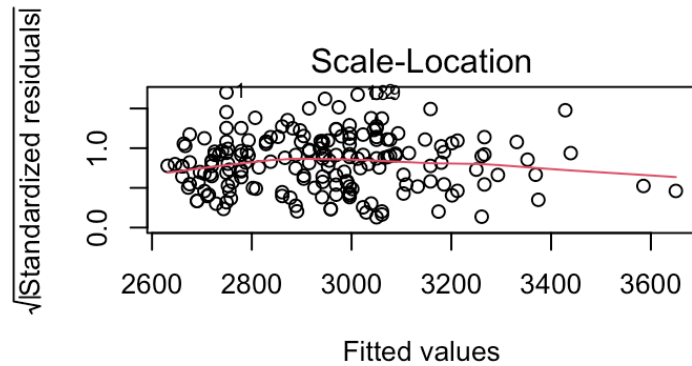
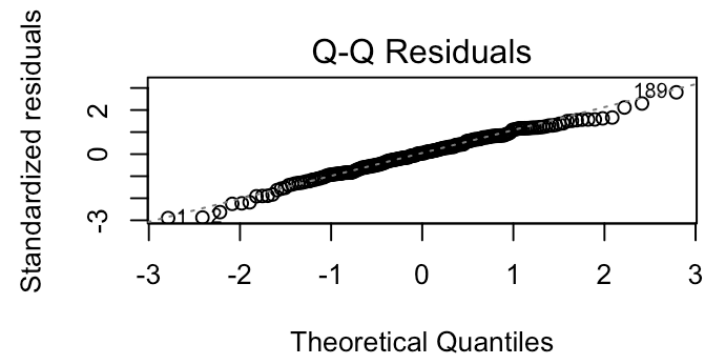
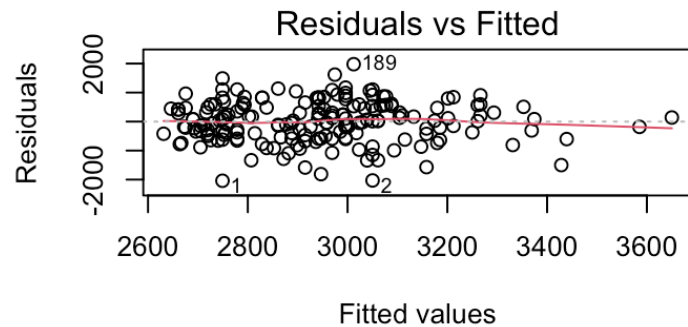
Fitted model in R

```
##
## Call:
## lm(formula = BWT ~ LWT * SMK, data = birth, contrasts = "contr.treatment")
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2040.25  -456.20   29.07   531.72  1977.72
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2347.507    312.717   7.507 2.48e-12 ***
## LWT           5.405      2.335   2.315  0.0217 *
## SMK(YES)      47.867    451.163   0.106  0.9156
## LWT:SMK(YES)  -2.456      3.388  -0.725  0.4695
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## s: 708.7 on 185 degrees of freedom
## Multiple R-squared: 0.06995,
## Adjusted R-squared: 0.05487
## F-statistic: 4.638 on 3 and 185 DF,  p-value: 0.003756
```

The estimated model(s)



Model summaries, tests, diagnostics



Model summaries, tests, diagnostics

You should be able to check:

- Are model assumptions met?
- Are there significant effects of LWT and/or SMK?
- How is the model fit in terms of R^2 ? Criticise the model!